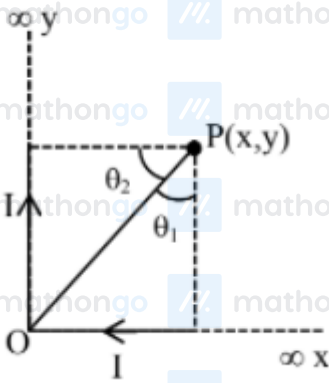


Q1 2021 (01 Sep Shift 2)

There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductor is 1 : 1. The magnetic field at point P is



- (1) $\frac{\mu_0 I}{4\pi xy} \left[\sqrt{x^2 + y^2} + (x + y) \right]$
- (2) $\frac{\mu_0 I}{4\pi xy} \left[\sqrt{x^2 + y^2} - (x + y) \right]$
- (3) $\frac{\mu_0 Ixy}{4\pi} \left[\sqrt{x^2 + y^2} - (x + y) \right]$
- (4) $\frac{\mu_0 Ixy}{4\pi} \left[\sqrt{x^2 + y^2} + (x + y) \right]$

Q2 2021 (31 Aug Shift 2)

A long solenoid with 1000 turns/m has a core material with relative permeability 500 and volume 10^3 cm^3 . If the core material is replaced by another material having relative permeability of 750 with same volume maintaining same current of 0.75 A in the solenoid, the fractional change in the magnetic moment of the core would be approximately $\left(\frac{x}{499} \right)$. Find the value ofx.

Q3 2021 (31 Aug Shift 2)

A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is :

(Assume that the current is flowing in the clockwise direction.)

- (1) $3 \times 10^{-7} \text{ T}$, outside the plane of triangle

(2) $2\sqrt{3} \times 10^{-7}$ T, outside the plane of triangle

(3) $2\sqrt{3} \times 10^{-5}$ T, inside the plane of triangle

(4) 3×10^{-5} T, inside the plane of triangle

Q4 2021 (31 Aug Shift 1)

A coil having N turns is wound tightly in the form of a spiral with inner and outer radii ' a ' and ' b ' respectively. Find the magnetic field at centre, when a current I passes through coil:

(1) $\frac{\mu_0 IN}{2(b-a)} \log_e \left(\frac{b}{a} \right)$

(2) $\frac{\mu_0 I}{8} \left[\frac{a+b}{a-b} \right]$

(3) $\frac{\mu_0 I}{4(a-b)} \left[\frac{1}{a} - \frac{1}{b} \right]$

(4) $\frac{\mu_0 I}{8} \left(\frac{a-b}{a+b} \right)$

Q5 2021 (27 Aug Shift 2)

A coaxial cable consists of an inner wire of radius ' a ' surrounded by an outer shell of inner and outer radii ' b ' and ' c ' respectively. The inner wire carries an electric current i_0 , which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) $x < a$ and (ii) $a < x < b$?

(1) $\frac{x^2}{a^2}$

(2) $\frac{a^2}{x^2}$

(3) $\frac{x^2}{b^2 - a^2}$

(4) $\frac{b^2 - a^2}{x^2}$

Q6 2021 (27 Aug Shift 1)

A uniform conducting wire of length is $24a$, and resistance R is wound up as a current carrying coil in the shape of an equilateral triangle of side ' a ' and then in the form of a square of side ' a '. The coil is connected to a voltage source V_0 . The ratio of magnetic moment of the coils in case of equilateral triangle to that for square is $1 : \sqrt{y}$ where y is

Q7 2021 (27 Aug Shift 1)

Two ions of masses 4amu and 16 amu have charges $+2e$ and $+3e$ respectively. These ions pass through the region of constant perpendicular magnetic field. The kinetic energy of both ions is same. Then :

- (1) lighter ion will be deflected less than heavier ion
- (2) lighter ion will be deflected more than heavier ion
- (3) both ions will be deflected equally
- (4) no ion will be deflected.

Q8 2021 (26 Aug Shift 2)

A coil in the shape of an equilateral triangle of side 10 cm lies in a vertical plane between the pole pieces of permanent magnet producing a horizontal magnetic field 20mT. The torque acting on the coil when a current of 0.2 A is passed through it and its plane becomes parallel to the magnetic field will be $\sqrt{x} \times 10^{-5} \text{Nm}$. The value of x is.

Q9 2021 (26 Aug Shift 2)

If the maximum value of accelerating potential provided by a radio frequency oscillator is 12kV. The number of revolution made by a proton in a cyclotron to achieve one sixth of the speed of light is

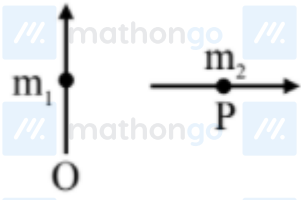
$$[m_p = 1.67 \times 10^{-27} \text{ kg}, e = 1.6 \times 10^{-19} \text{ C}]$$

$$\text{Speed of light} = 3 \times 10^8 \text{ m/s}]$$

Q10 2021 (26 Aug Shift 1)

Two short magnetic dipoles m_1 and m_2 each having magnetic moment of 1Am^2 are placed at point O and P respectively. The distance between OP is 1 meter. The torque experienced by the magnetic dipole m_2 due to

the presence of m_1 is $\times 10^{-7} \text{ Nm}$



Q11 2021 (26 Aug Shift 1)

The fractional change in the magnetic field intensity at a distance 'r' from centre on the axis of current carrying coil of radius 'a' to the magnetic field intensity at the centre of the same coil is: (Take $r < a$)

- (1) $\frac{3}{2} \frac{a^2}{r^2}$
- (2) $\frac{2}{3} \frac{a^2}{r^2}$
- (3) $\frac{2}{3} \frac{r^2}{a^2}$
- (4) $\frac{3}{2} \frac{r^2}{a^2}$

Answer Key

Q1 (1)

Q2 (250)

Q3 (4)

Q4 (1)

Q5 (1)

Q6 (3)

Q7 (2)

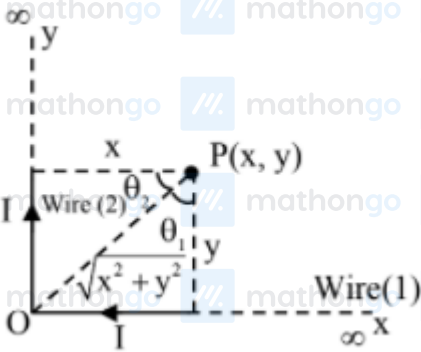
Q8 (3)

Q9 (543)

Q10 (1)

Q11 (4)

Q1 (1)



$$B_{\text{due to wire (1)}} = \frac{\mu_0 I}{4\pi y} [\sin 90^\circ + \sin \theta_1]$$

$$= \frac{\mu_0 I}{4\pi y} \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right) \dots (1)$$

$$B_{\text{due to wire (2)}} = \frac{\mu_0 I}{4\pi x} (\sin 90^\circ + \sin \theta_2)$$

$$= \frac{\mu_0 I}{4\pi x} \left(1 + \frac{y}{\sqrt{x^2 + y^2}} \right) \dots (2)$$

Total magnetic field

$$B = B_1 + B_2 \quad B = \frac{\mu_0 I}{4\pi} \left[\frac{1}{y} + \frac{x}{y\sqrt{x^2 + y^2}} + \frac{1}{x} + \frac{y}{x\sqrt{x^2 + y^2}} \right] \quad B = \frac{\mu_0 I}{4\pi} \left[\frac{x+y}{xy} + \frac{x^2 + y^2}{xy\sqrt{x^2 + y^2}} \right]$$

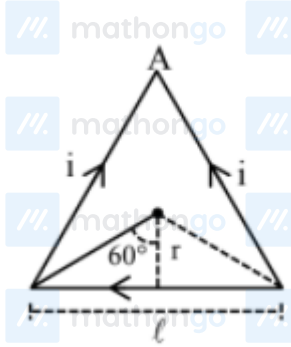
$$B = \frac{\mu_0 I}{4\pi} \left[\frac{x+y}{xy} + \frac{\sqrt{x^2 + y^2}}{xy} \right] \quad B = \frac{\mu_0 I}{4\pi xy} [\sqrt{x^2 + y^2} + (x + y)] \quad \text{Option (1)}$$

Q2 (250)

$$\frac{\Delta M}{M} = \frac{\Delta \mu}{\mu} = \frac{250}{500} = \frac{1}{2}$$

$$\frac{1}{2} = \frac{x}{499} \Rightarrow x \simeq 250$$

Q3 (4)



$$B = 3 \left[\frac{\mu_0 i}{4\pi r} (\sin 60^\circ + \sin 60^\circ) \right]$$

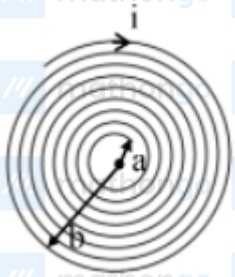
$$\tan 60^\circ = \frac{\ell/2}{r}$$

$$\text{Where } r = \frac{9 \times 10^{-2}}{2\sqrt{3}} \text{ m}$$

$$\therefore B = 3 \times 10^{-5} \text{ T}$$

Current is flowing in clockwise direction so, $\vec{B}$ is inside plane of triangle by right hand rule.

Q4 (1)



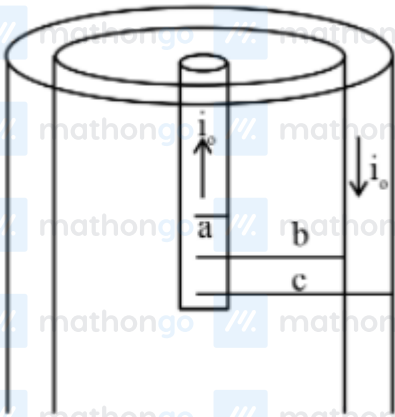
$$\text{No. of turns in } dx \text{ width} = \frac{N}{b-a} dx$$

$$\int dB = \int_a^b \left(\frac{N}{b-a} \right) dx \frac{\mu_0 i}{2x}$$

$$B = \frac{N\mu_0 i}{2(b-a)} \ln \left(\frac{b}{a} \right)$$

Option (1)

Q5 (1)

when $x < a$

$$B_1(2\pi x) = \mu_0 \left(\frac{i_0}{\pi a^2} \right) \pi x^2$$

$$B(2\pi x) = \frac{\mu_0 i_0 x^2}{a^2}$$

$$B_1 = \frac{\mu_0 i_0 x}{2\pi a^2}$$

when $a < x < b$

$$B_2(2\pi x) = \mu_0 i_0$$

$$B_2 = \frac{\mu_0 i_0}{2\pi x}$$

$$\frac{B_1}{B_2} = \frac{\frac{\mu_0 i_0 x}{2\pi a^2}}{\frac{\mu_0 i_0}{2\pi x}} = \frac{x^2}{a^2}$$

Q6 (3)

$$\text{In triangle shape } N_t = \frac{24a}{3a} = 8$$

$$\text{In square } N_s = \frac{24a}{4a} = 6$$

$$\frac{M_t}{M_s} = \frac{N_t I A_t}{N_s I A_s} \text{ [I will be same in both]}$$

$$= \frac{8 \times \frac{\sqrt{3}}{4} \times a^2}{6 \times a^2}$$

$$\frac{M_t}{M_s} = \frac{1}{\sqrt{3}}$$

$$y = 3$$

Q7 (2)

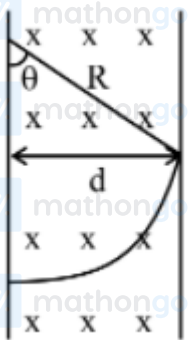
$$r = \frac{P}{qB} = \frac{\sqrt{2mk}}{qB}$$

Given they have same kinetic energy

$$r \propto \frac{\sqrt{m}}{q}$$

$$\frac{r_1}{r_2} = \frac{\sqrt{4}}{2} \times \frac{3}{\sqrt{16}} = \frac{3}{4}$$

$$r_2 = \frac{4r_1}{3} \text{ (} r_2 \text{ is for heavier ion and } r_1 \text{ is for lighter ion)}$$



$$\sin \theta = \frac{d}{R}$$

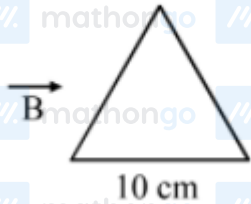
 $\theta \rightarrow$ Deflection

$$\theta \propto \frac{1}{R}$$

($R \rightarrow$ Radius of path)

$$\because R_2 > R_1 \Rightarrow \theta_2 < \theta_1$$

Q8 (3)



$$\begin{aligned}\vec{\tau} &= \vec{M} \times \vec{B} = MB \sin 90^\circ \\ &= MB = \frac{i\sqrt{3}\ell^2}{4} B \\ &= \sqrt{3} \times 10^{-5} \text{ N-m}\end{aligned}$$

Ans. 3

Q9 (543)

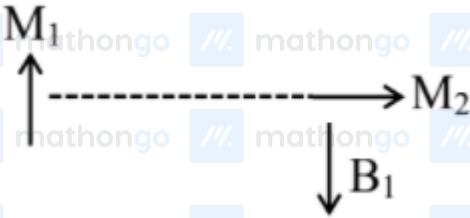
$$V = 12\text{kV}$$

Number of revolution = n

$$\begin{aligned}n [2 \times q_p \times V] &= \frac{1}{2} m_p \times v_p^2 \\ n [2 \times 1.6 \times 10^{19} \times 12 \times 10^3] &= \frac{1}{2} \times 1.67 \times 10^{-27} \times \left[\frac{3 \times 10^8}{6} \right]^2 \\ n (38.4 \times 10^{-16}) &= 0.2087 \times 10^{-11} \\ n &= 543.4\end{aligned}$$

Ans. 543

Q10 (1)



$$\vec{\tau} = \mathbf{M}_2 \times \mathbf{B}_1$$

$$\tau = M_2 B_1 \sin 90^\circ$$

$$= 1 \times \frac{\mu_0}{4\pi} \frac{M_1}{(1)^3} 1$$

$$= 10^{-7} \text{ N} \cdot \text{m}$$

Ans. 1.00

Q11 (4)

$$B_{\text{axis}} = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}$$

$$B_{\text{centre}} = \frac{\mu_0 i}{2R}$$

$$\therefore B_{\text{centre}} = \frac{\mu_0 i}{2a}$$

$$\therefore B_{\text{axis}} = \frac{\mu_0 i a^2}{2(a^2 + r^2)^{3/2}}$$

$\therefore$ fractional change in magnetic field =

$$\frac{\mu_0 i}{2a} - \frac{\mu_0 i a^2}{2(a^2 + r^2)^{3/2}} \frac{\mu_0 i}{2a} = 1 - \frac{1}{\left[1 + \left(\frac{r^2}{a^2}\right)\right]^{3/2}}$$

$$2] - \left[1 - \frac{3}{2} \frac{r^2}{a^2}\right] = \frac{3}{2} \frac{r^2}{a^2}$$

Note : $\left(1 + \frac{r^2}{a^2}\right)^{-3/2} \approx \left(1 - \frac{3}{2} \frac{r^2}{a^2}\right)$

[True only if $r \ll a$]

Hence option (4) is the most suitable option